

National Real-Time Payments System – Business Report

1. Introduction

This report presents a clear and engaging view of the National Real-Time Payments System. It explains why the system is important for the country, what it will deliver, and how it supports the long-term growth of digital payments. The report focuses more on business understanding than technical design.

2. Purpose and Vision

The main purpose of the National Real-Time Payments System is to help people and businesses send and receive money instantly. Today, many transactions depend on cash or slow banking transfers. This new system provides a faster, safer, and more modern way to transfer money. It also supports the national goal of reducing cash usage and improving financial transparency.

The long-term vision is to make digital payments simple and accessible for everyone—citizens, merchants, banks, and government bodies. The system will become a strong foundation for the country's payment ecosystem and will help enable future cross-border international payments as well.

3. Key User Groups

The system will be used by several important user groups:

- Citizens who make quick payments to friends, family, or service providers at any time.
- Merchants who want a convenient and safe way to accept money.
- Banks and financial institutions that process and manage transactions.
- Regulators and auditors who ensure fairness, trust, and compliance across the ecosystem.

4. Core Business Capabilities

The system provides several important capabilities that support smooth and reliable payments. These include:

- Easy and secure onboarding for users and merchants.
- A feature that lets users link and manage multiple bank accounts.
- Instant payments for both P2P (person-to-person) and P2M (person-to-merchant).

- Ability for merchants to generate and use QR codes for payments.
- Clear transaction history and status updates for users.
- Automated settlement and reconciliation for banks.
- Support for dispute management and fraud checks to maintain trust.

5. Technical Architecture

The NRTP system uses modular and layered architecture. Each layer handles a specific function to keep the system fast, scalable, and easy to maintain. The architecture is explained in simple terms below:

- **API Layer** – Handles all incoming requests from mobile apps, web apps, and merchant systems.
- **Authentication Service** – Verifies user identity, bank account links, and authorization requests.
- **Payment Orchestrator** – Manages the complete payment journey including validations, routing, and transaction flow.
- **Realtime Ledger** – Records every transaction immediately to ensure accuracy, traceability, and no data loss.
- **Fraud & Risk Engine** – Runs instant risk rules and asynchronous machine-learning checks.
- **Bank Connector Layer** – Connects to every bank for account checks, balance verification, and settlement messages.
- **Monitoring & Alerts** – Tracks system health, latency, errors, and performance indicators in real time.

6. How the Payment Process Works (Technical Overview)

The following explains the flow of a typical payment using simple technical language:

1. The user or merchant starts a payment via app or QR code.
2. The API Layer receives the request and checks authentication.
3. The Payment Orchestrator validates the details and checks the fraud engine.
4. If everything is safe, the system sends a request to the user's bank to confirm funds.
5. Once the bank confirms, the system records the transaction in the real-time ledger.
6. A confirmation message is sent to both sender and receiver.
7. Banks complete settlements either instantly or at scheduled intervals.

7. Business Benefits

The system brings many advantages to the national economy and financial ecosystem. These benefits support growth, trust, and better customer experiences.

- Faster payments improve customer satisfaction and encourage digital adoption.
- Merchants gain faster access to funds, improving their cash flow and planning.
- Banks gain a reliable and standardised payment platform.
- Government agencies benefit from better reporting through digital transaction records and reduced cash-related risks.
- The overall financial system becomes more transparent and efficient.

8. Service Expectations (Non-functional Requirements)

The system must perform well even under high demand. These service expectations ensure the platform is fast, secure, and dependable.

- Payments must be completed in less than 2 seconds.
- The system should be available 99.999% of the time.
- It must support up to 50,000+ transactions per second.
- It must manage up to 10 times the normal traffic during national peak days.
- Data must be kept safe, secure and encrypted at all stages.
- There should be no single point of failure in the system design.

9. Future Expansion Opportunities

The system is built with long-term expansion in mind. As digital payments grow worldwide, the platform can support new features and global connections. This includes:

- Support for sending and receiving payments across borders.
- Handling multiple currencies smoothly.
- Integration with foreign banks and digital platforms.
- Compliance with international financial rules and standards.

10. Conclusion

The National Real-Time Payments System is a major step towards a modern and trusted digital economy. It supports citizens, businesses, banks, and public institutions with faster payments and safer processes. By planning for future global connections, the system also helps the country stay competitive in a fast-changing world. This platform is not just a technical solution—it is a long-term investment in the nation's digital future.